CCSA104 PRACTICAL ASSIGNMENT: MANAGING SECURITY POLICIES

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Course: CCSA104 Security Policy Management
Date completed: 8 Nov 2026

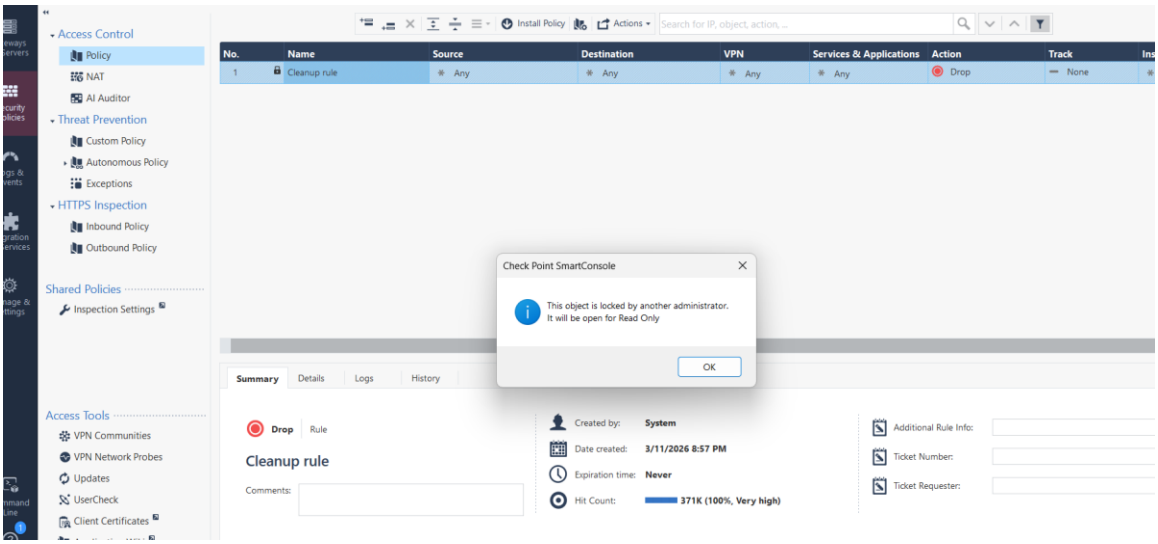
Limitations and Issue Log

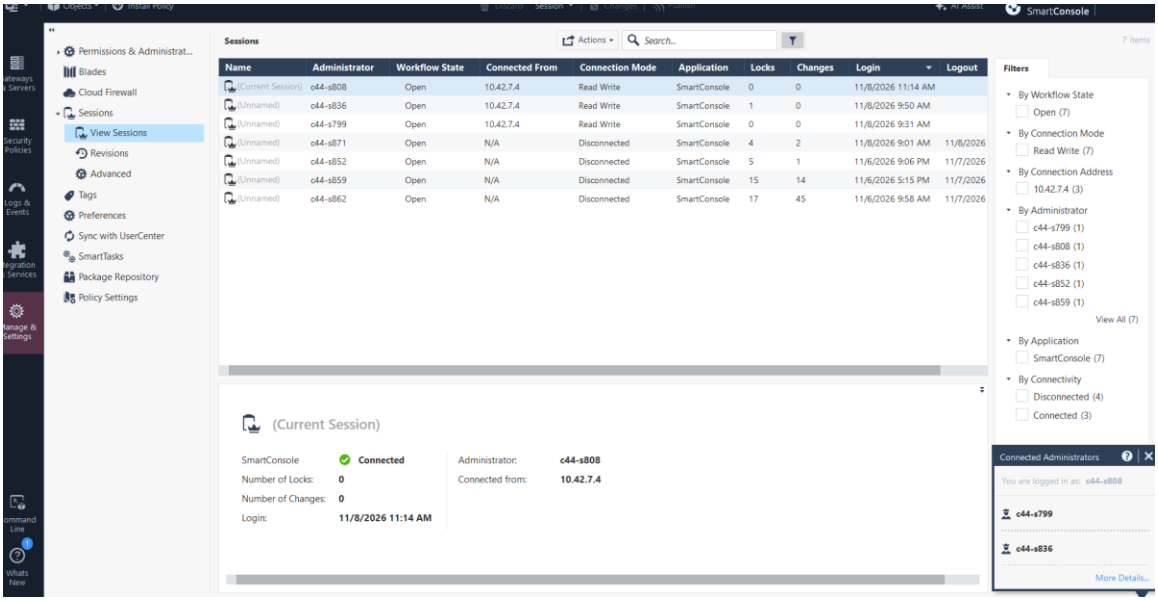
#	Issue	Impact	Action taken
1	The brief I received was numbered 1, 2, 4, 6, 7, with no Task 3 or Task 5.	Task 2 says rule 4 ("Internal to DMZ") is "added in Task 5", so its values were not given.	Emailed ICDFA Student Support at 12:17 AM on 6 Nov 2026 and messaged instructor Udam Akume Gabriel on Teams at 12:20 AM on 6 Nov 2026. He replied on Teams at 5:38 AM on 7 Nov 2026 that the numbering was an oversight and the tasks should be reordered, with Task 4 becoming Task 3. I renumbered this report to match: sections are Task 3, Hide NAT is Task 4, and the rule order experiment is Task 5. His reply did not give the rule 4 values or test T4. I followed up on Teams at 9:14 AM on 7 Nov 2026 about rule 4, test T4 and the demonstration. No reply as of 8:07 PM on 8 Nov 2026.
2	The rule order task (Task 7 in my copy) says to "repeat test T4", but no test T4 is defined in my copy of the brief.	I could not follow a defined test.	Same request sent to the instructor. No substitute test was run, because Task 5 could not be performed (issue 5).
3	The portal submission page requires an instructor-verified live or recorded demonstration. The brief does not say what it should show.	Unclear demonstration scope.	Asked the instructor in the same message. No reply as of 10:07 PM on 8 Nov 2026.
4	Cloud lab policy locked. On 8 Nov 2026 at about 11:15 AM SmartConsole said the object is locked	I could not edit the cloud policy, so Tasks 2 to 5	Sent a Teams message with screenshots to the instructor at 11:30 AM on 8 Nov 2026, asking him to clear the stale sessions, advise on name clashes, and advise how to install on

#	Issue	Impact	Action taken
	by another administrator and opened the Standard policy read-only. The Sessions view showed 7 sessions on the shared server. Four disconnected sessions still held locks and unpublished changes, and my session had none.	were blocked there.	the shared gateway. With no reply and the deadline at 11:59 PM that night, I built Tasks 2 to 4 on my local A-SMS.
5	Task 5 (rule order experiment) could not be performed. It needs a policy installation on a Security Gateway and traffic logs.	My local lab has no gateway VM, and the cloud policy was locked with no instructor reply, so there is no installation or log evidence.	Not attempted, and no substitute evidence was created. The explanation under Task 5 is theory only, and Part B Q7 also covers rule order.

Assumption for rule 4 (Internal to DMZ): because no task in the brief gives the values for rule 4 (Task 2 says it is added in Task 5, but the renumbered Task 5 is the rule order experiment), I built this rule from my own judgement: Source Internal_LAN, Destination DMZ_Net, Services http and https, Action Accept, Track Log. It is not based on instructions from the brief.

Issue 4 evidence (cloud lab, 8 Nov 2026): SmartConsole's lock message, and the Sessions view with 7 sessions on the shared server. My session (current) has 0 locks and 0 changes.





Part A: Practical Tasks

Task 1: Explore the default policy

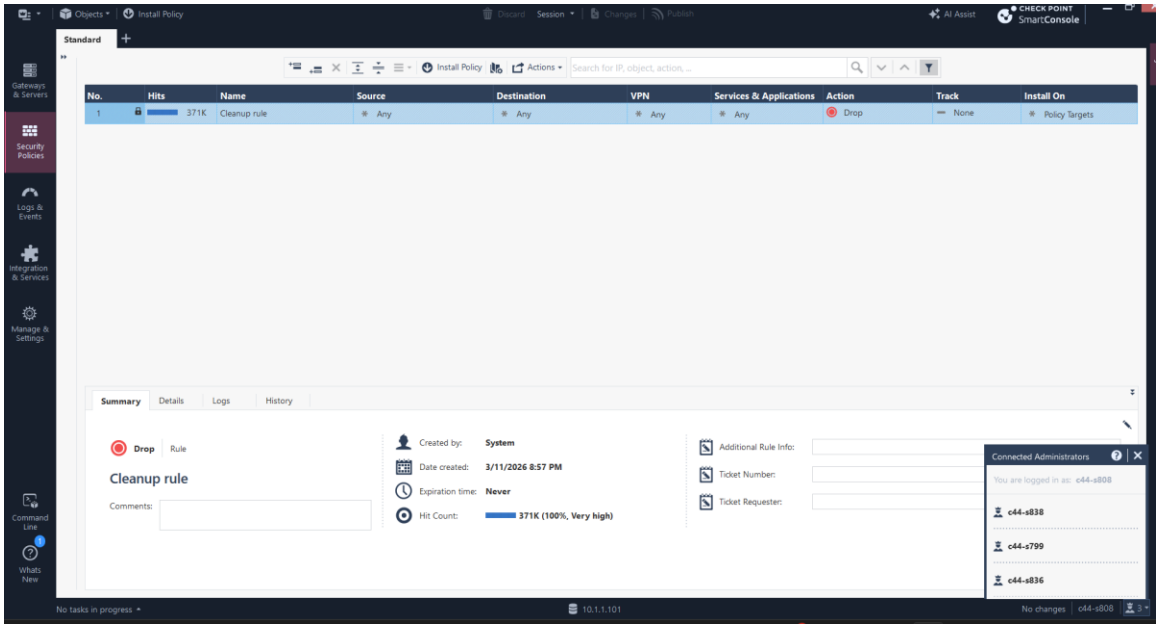
Policy package in use: Standard (Access Control and Threat Prevention, Policy Targets: All gateways)

Cleanup rule field	Value
Name	Cleanup rule
Source	Any
Destination	Any
Services & Applications	Any
Action	Drop
Track	None
Install On	Policy Targets

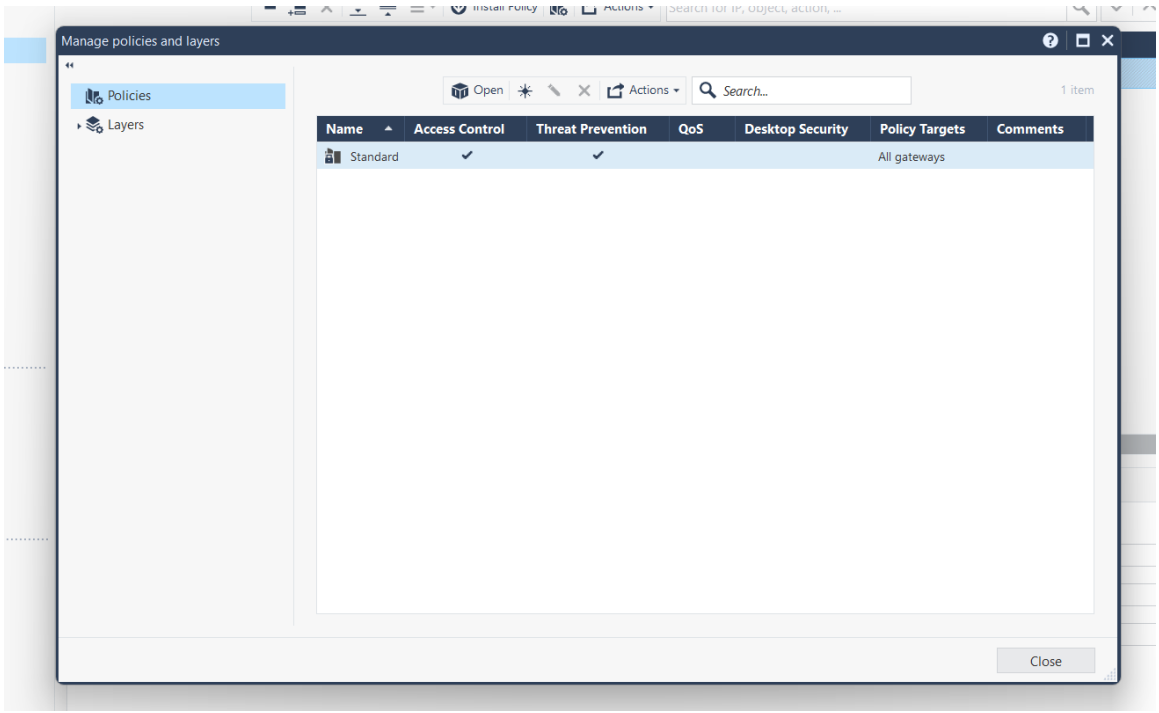
Global Properties categories seen (three): Firewall, VPN, Log and Alert

Policy packages listed under Manage policies and layers: Standard is the only policy package. Under Layers, Access Control has Network, Threat Prevention has IPS and Standard Threat Prevention, and HTTPS Inspection has Default Inbound Layer and Default Outbound Layer.

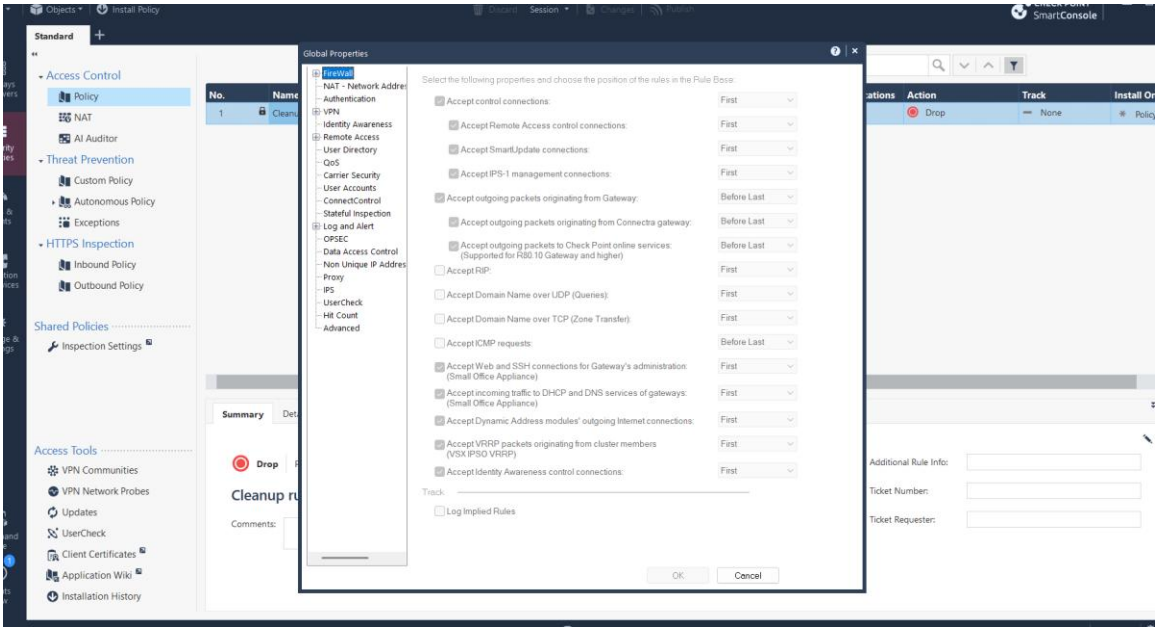
Evidence: Task 1 - Default rulebase with Hits, Track and Install On columns visible (cloud lab, 10.1.1.101, account c44-s808).



Evidence: Task 1 - Manage policies and layers, Policies list (one package, Standard).

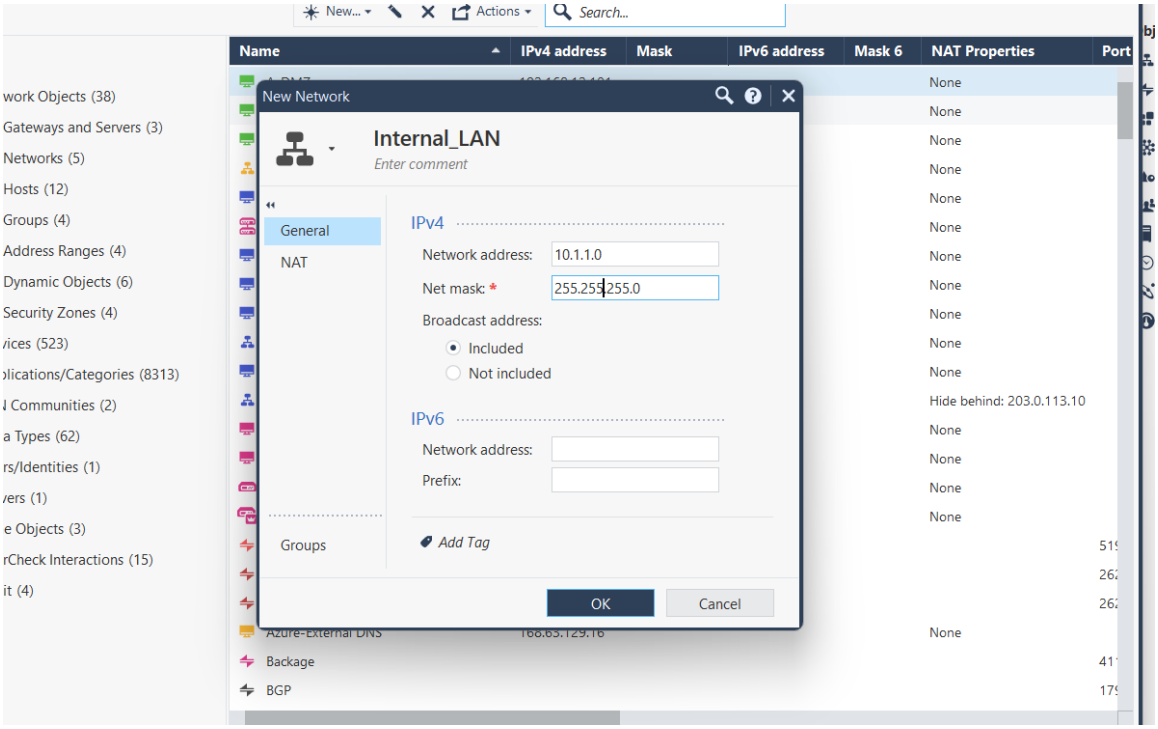


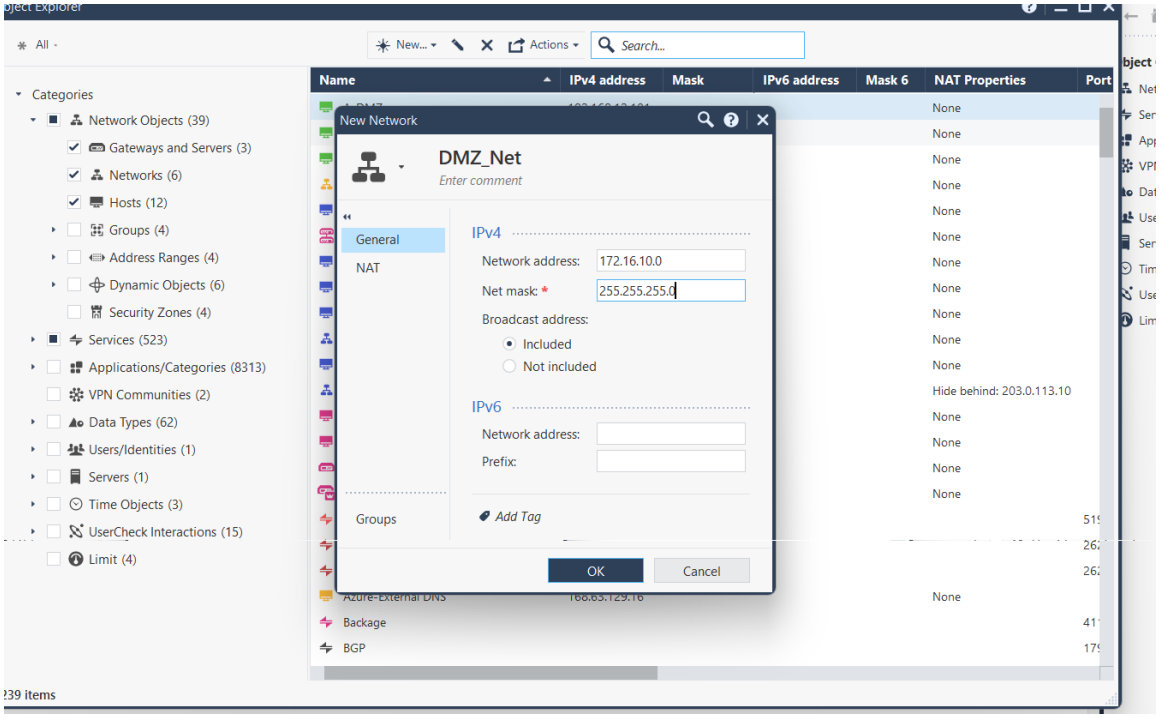
Evidence: Task 1 - Global Properties category list (opened and cancelled, nothing changed).

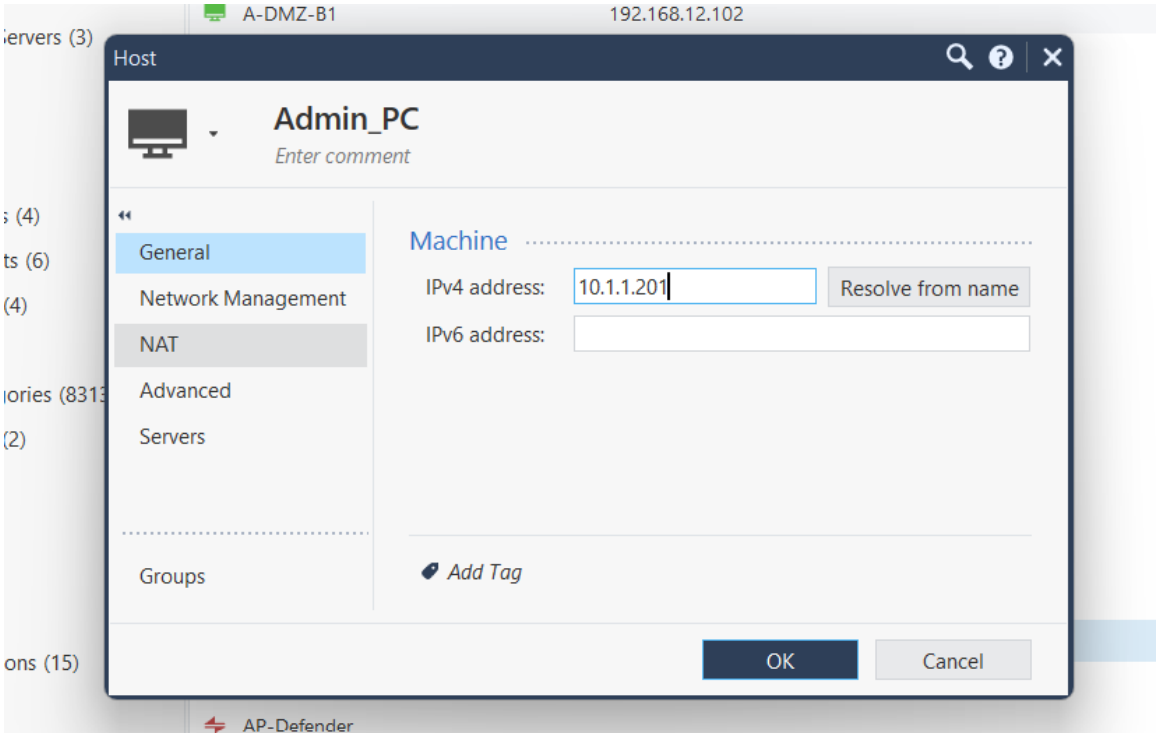


Task 2: Build the Access Control rulebase

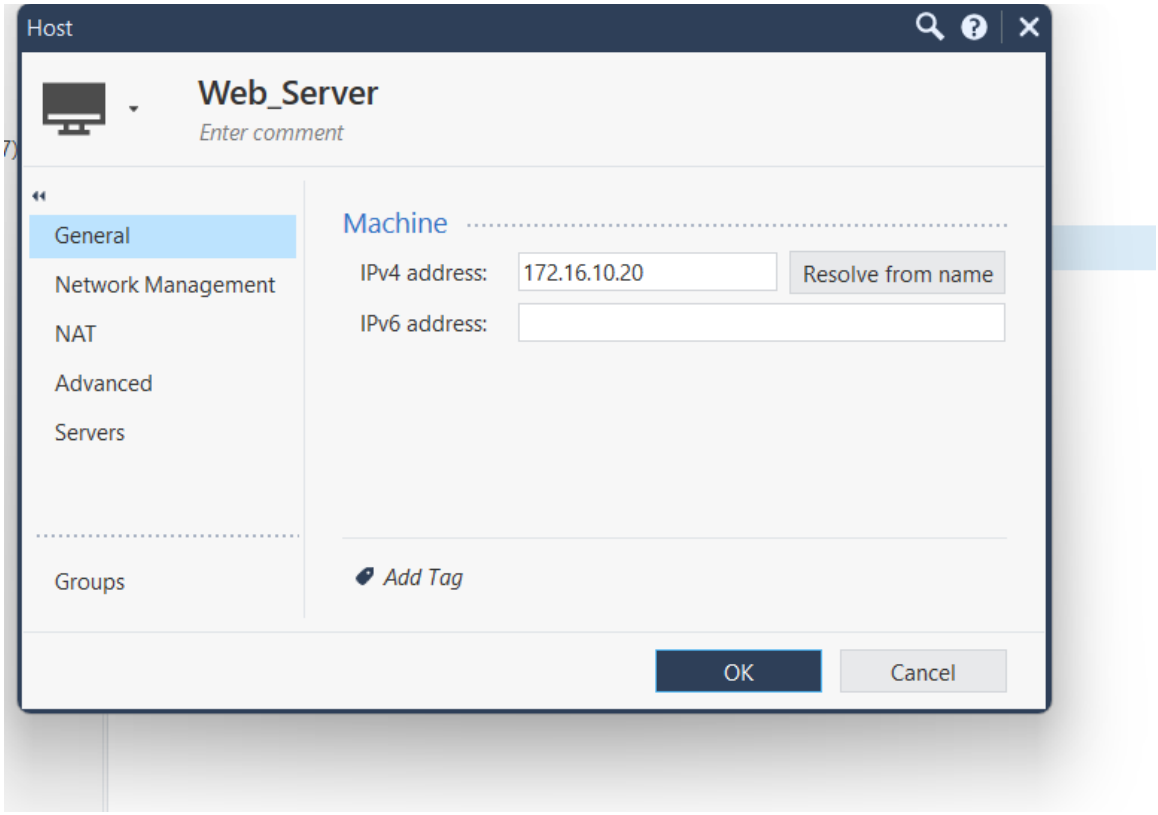
Objects created: Internal_LAN (10.1.1.0/24), DMZ_Net (172.16.10.0/24), Admin_PC (10.1.1.201, my workstation's VMnet8 address), Web_Server (172.16.10.20), Mail_Server (172.16.10.25) and External_DNS (192.9.100.10). Tasks 2 to 4 were done on my local A-SMS (account cpadmin, 10.1.1.101) because the cloud policy was locked (see issue 4). The gateway object used is A-GW-Cluster.

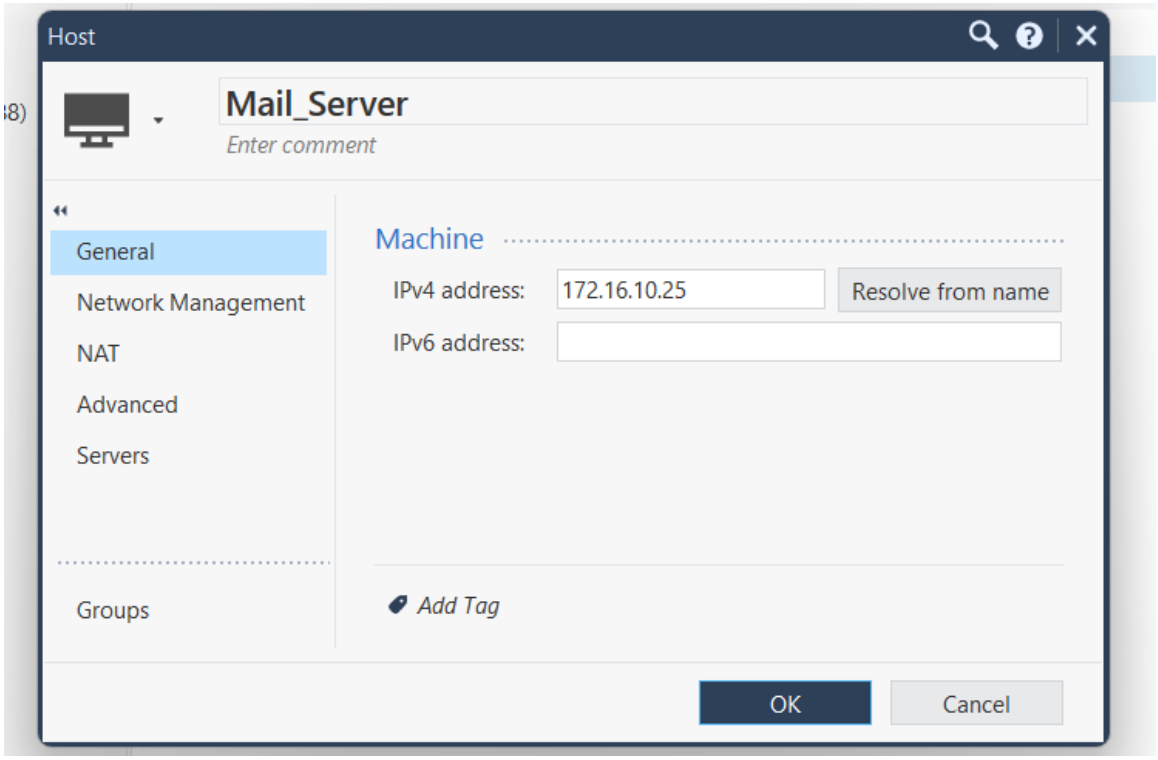


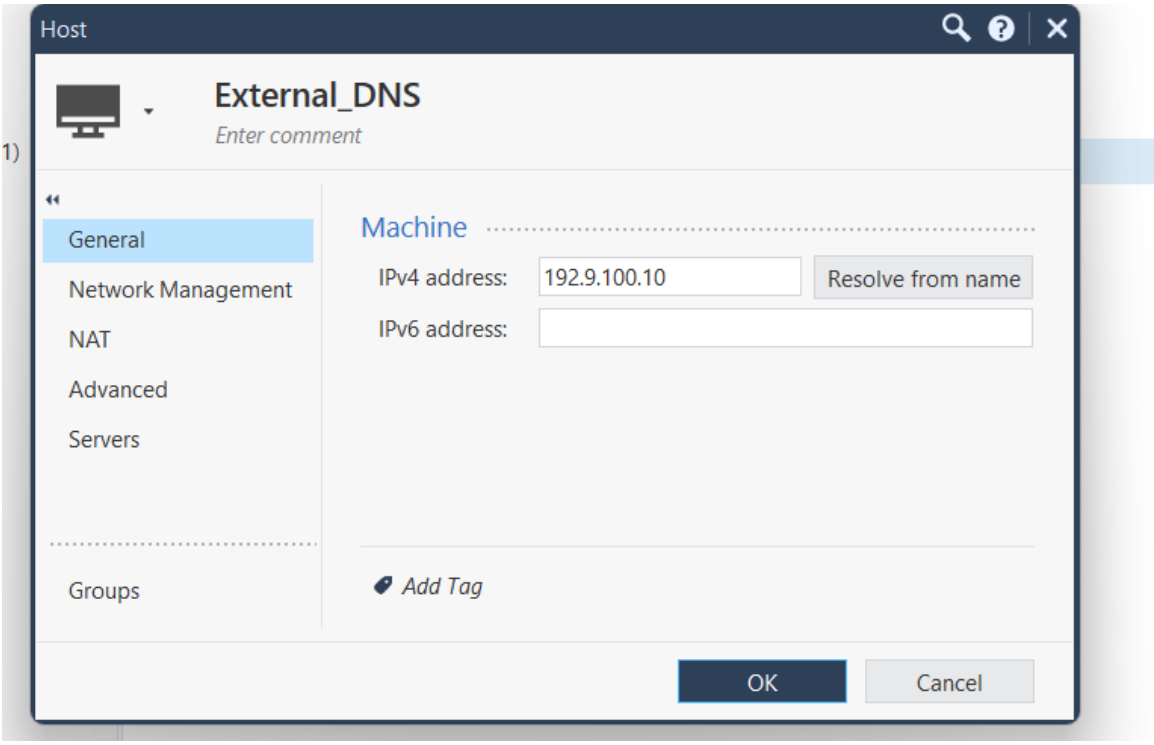




Admin_PC uses 10.1.1.201 because that is my workstation's address on the lab network, shown in ipconfig below.

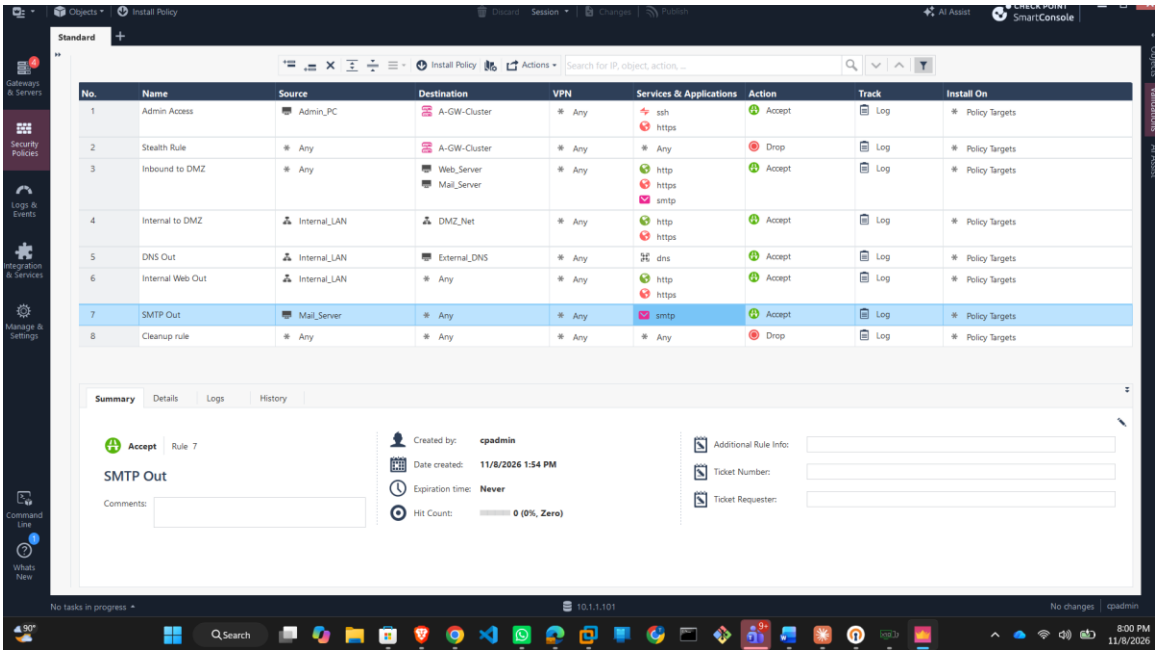

Host
Web_Server
Enter comment
General
Network Management
NAT
Advanced
Servers
Groups
Machine
IPv4 address:
172.16.10.20
Resolve from name
IPv6 address:
Add Tag
OK
Cancel


Host
Mail_Server
Enter comment
General
Network Management
NAT
Advanced
Servers
Groups
Machine
IPv4 address:
172.16.10.25
Resolve from name
IPv6 address:
Add Tag
OK
Cancel



Cleanup Rule: Action set to Drop and Track set to Log, and left at the bottom of the rulebase. On my local A-SMS it started as Accept.

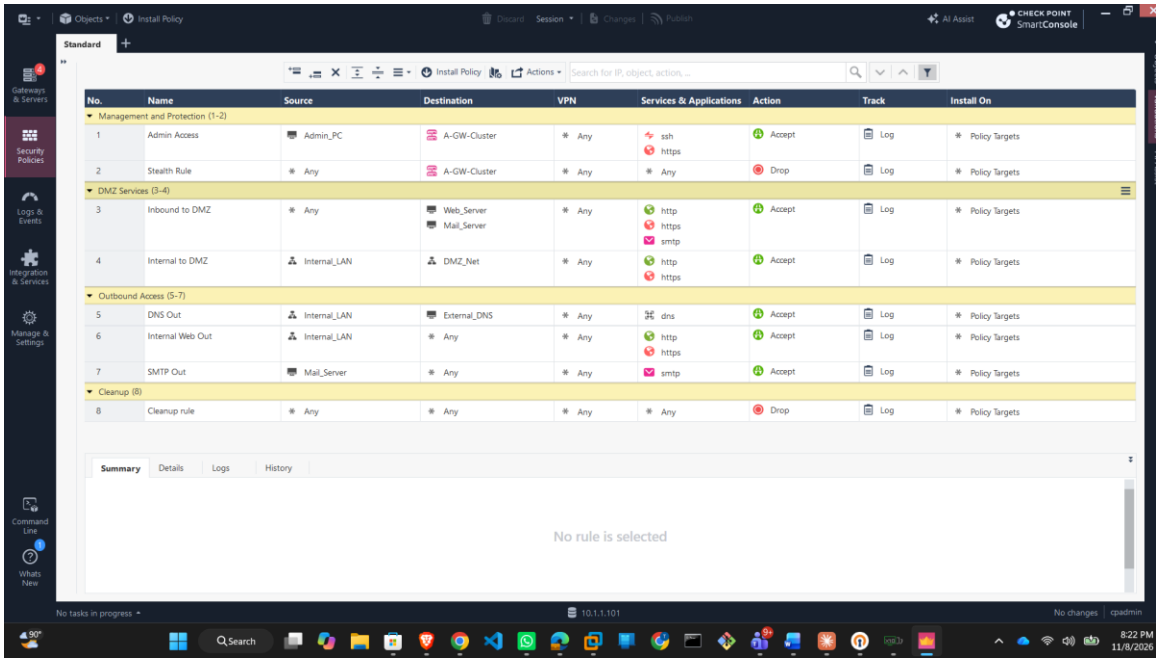
Evidence: Task 2 - Full rulebase showing No., Name, Source, Destination, Services, Action and Track (local A-SMS, cpadmin, 10.1.1.101, published).



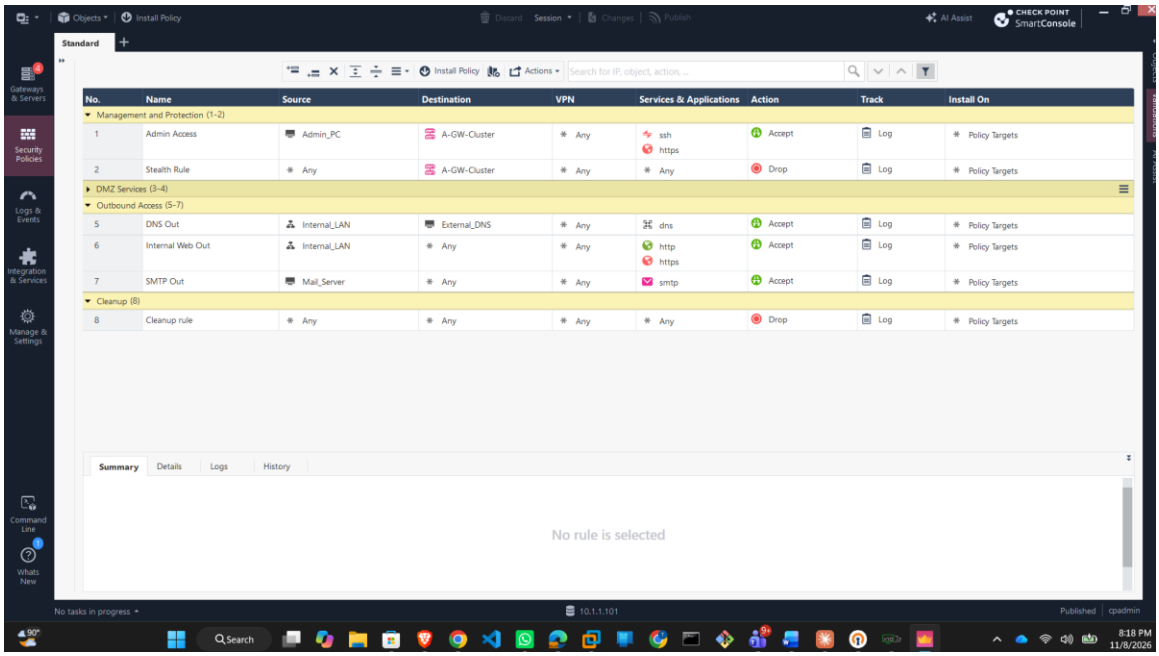
Task 3: Organise the rulebase with sections (Task 4 in the original brief)

Sections created: Management and Protection (rules 1-2), DMZ Services (rules 3-4), Outbound Access (rules 5-7), Cleanup (rule 8). I collapsed and expanded a section.

Evidence: Task 3 - All sections expanded (published).



Evidence: Task 3 - One section collapsed (DMZ Services).

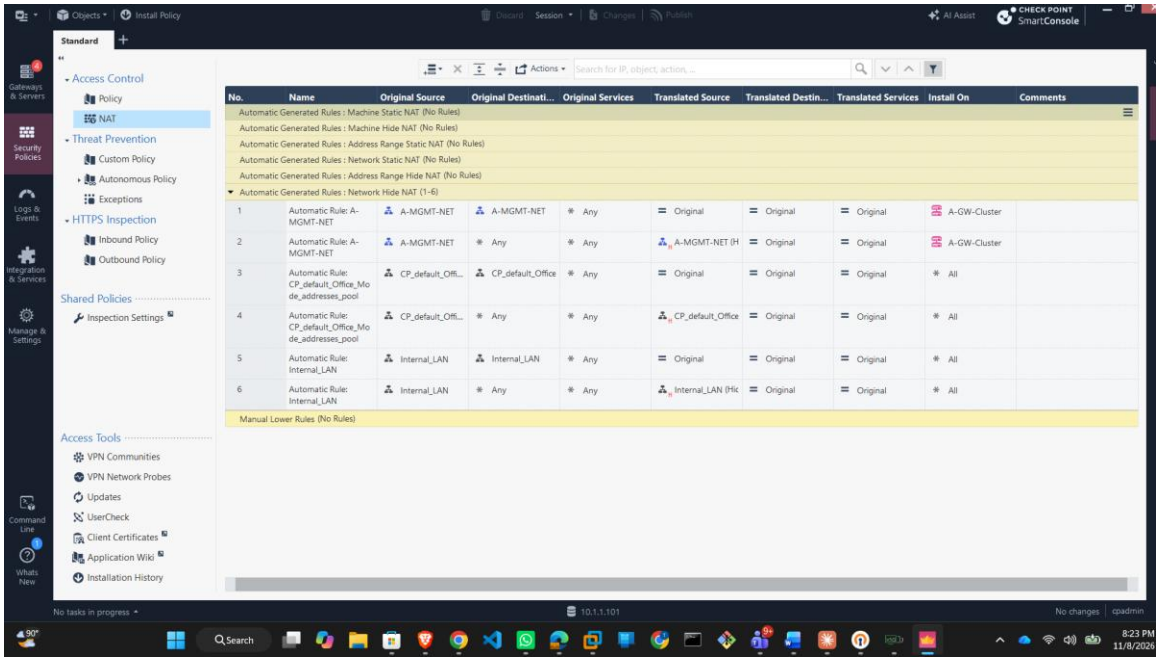


Statement: Sections are only a visual way to group rules. They do not change the order in which rules are enforced, and they are not sent to the Security Gateway as part of the enforcement logic.

Task 4: Configure Hide NAT (Task 6 in the original brief)

On Internal_LAN, NAT page: automatic address translation enabled, Hide selected, hidden behind the gateway. Changes published.

Evidence: Task 4 - NAT rulebase with the automatic rules for Internal_LAN (rules 5 and 6), published.



Evidence: [Insert screenshot: Task 4 - Internal_LAN NAT settings]

Explanation: The rule is automatic because the Security Management Server generated it from the NAT settings I enabled on the Internal_LAN object, rather than me writing it in the NAT rulebase by hand. Many-to-one translation means many internal addresses share a single translated address, here the gateway's address. For each connection the gateway also changes the source port to a new one and records the mapping, so when the reply comes back it can match it to the correct internal host and translate the address back.

Task 5: Rule order experiment and Installation History (Task 7 in the original brief)

Task 5 was not performed. It needs a policy installation on a Security Gateway and traffic logs showing which rule matched. My local lab had no gateway VM, and the cloud policy was locked with no response from the instructor (issues 4 and 5). I therefore have no installation, log or Installation History evidence, and I have not created any.

Explanation (theory only, not from my own logs): Rules are checked from top to bottom and the first match decides what happens. If a broad temporary block rule for the internal network sat above Internal Web Out, it would match web traffic first and drop it, so the more specific rule would never be reached. Moving it below Internal Web Out would let the specific rule accept the traffic first. That is why specific rules belong above general ones, as in my rulebase where Admin Access sits above the Stealth Rule.

Part B: Theory Questions

Section B1: Multiple choice

Q1: B. Q2: A. Q3: C. Q4: D. Q5: B.

Section B2: Short answers

Q6. Explicit rules are created by the administrator, are visible in the rulebase, and allow or block traffic based on criteria the administrator sets. Implied rules are created in Global Properties on

the Security Management Server, are not shown in the rulebase, and can be placed first, last, or before the last explicit rule. The Implicit Cleanup Action is not a rule. It is gateway behaviour for traffic that matched nothing, and its options are Drop or Accept.

Q7. The gateway checks rules top to bottom and the first match applies, so specific rules go above general ones. In my rulebase, Admin Access (rule 1) is above the Stealth Rule (rule 2). If the Stealth Rule were first, it would drop my own SSH/HTTPS connection to the gateway. The Cleanup Rule is last so it only catches traffic that nothing else matched.

Q8. Saving is not the same as publishing. Edits in SmartConsole form a draft session on the management server, and publishing makes them official and visible to other administrators. Unpublished changes are not included when a policy is installed. Many organisations publish only in a change window so half-finished edits cannot reach production and changes can be reviewed and rolled back in a controlled way.

Q9. Global Properties are settings enforced by all gateways managed by the Security Management Server, covering areas such as Firewall, NAT, VPN, and Logging and Alerts; an example is the implied rules. They are opened from the Applications menu, then Global properties. A Policy Package is a logical grouping of policy types (Access Control, QoS, Desktop Security, Threat Prevention) that lets different combinations be installed on different gateways. It is opened from the Applications menu, then Manage policies and layers (Ctrl+O).

Q10. The four predefined zones are InternalZone, ExternalZone, DMZZone and WirelessZone. A gateway interface can belong to only one zone, and interfaces to different networks can share the same zone. A benefit of zone-based rules is that one rule covers every network in a zone, which keeps the rulebase shorter and easier to manage.

Section B3: Scenario questions

Q11. First, an Accept-any rule above the Stealth rule matches traffic aimed at the gateway too, so the Stealth rule never drops it and the gateway is directly exposed. Second, every rule below it, including the Cleanup rule, is never reached, so the policy effectively allows everything and breaks the Positive Control Model. To correct it, delete the rule or move it to the bottom, replace it with specific rules for only the traffic needed, and put the Stealth rule back near the top. Only Management rules (and other connections that must reach the gateway directly) may sit above the Stealth rule, because administrators must be able to connect to the gateway before the Stealth rule drops traffic to it.

Q12. (a) Two benefits of NAT: it hides internal IP addresses behind a public one, and it lets many devices share one public IP address, which conserves addresses.

(b) Three checks: (1) Confirm the change was published and installed, because unpublished changes are not included in an installation, so the gateway may not have the NAT rule; Installation History shows this. (2) Check the NAT rulebase for the automatic rule for Internal_LAN, with the right translated source and Install On gateway, because a missing or wrong translation means replies never return. (3) Check the logs for the traffic, because they show which rule matched: if the Cleanup rule is dropping it, an Access Control rule such as Internal Web Out or DNS Out is wrong or misplaced, and if it is accepted but still fails, routing or the external interface needs checking.

Declaration

I used Claude, an AI assistant from Anthropic, to troubleshoot lab connection problems. I carried out the lab steps and took the screenshots myself, and I reviewed the answers before submitting.